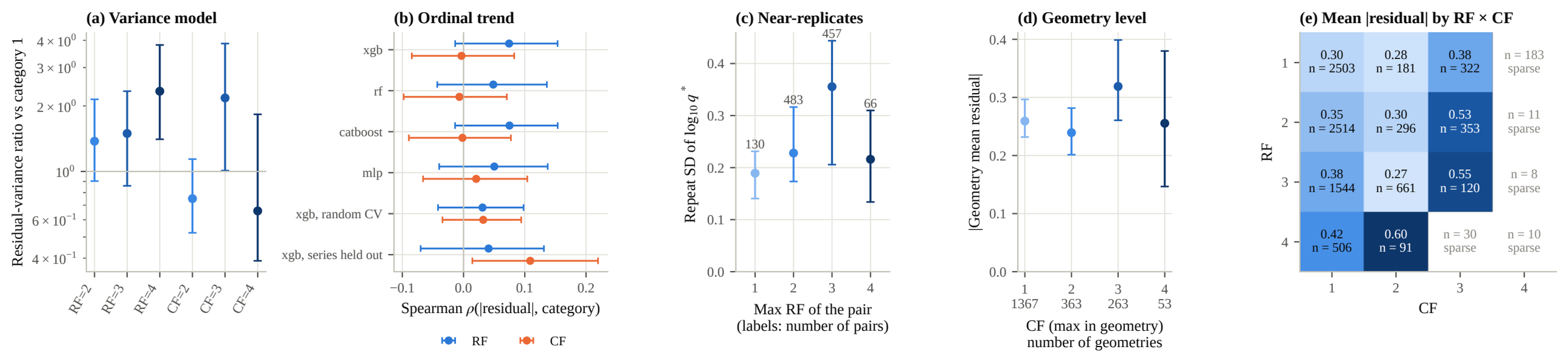
(a) Variance model
Residual-variance ratio vs category 1
4 × 10^0
3 × 10^0
2 × 10^0
10^0
6 × 10^−1
4 × 10^−1
RF=2
RF=3
RF=4
CF=2
CF=3
CF=4
(b) Ordinal trend
xgb
rf
catboost
mlp
xgb, random CV
xgb, series held out
−0.1
0.0
0.1
0.2
Spearman ρ(|residual|, category)
RF
CF
(c) Near-replicates
Repeat SD of log10 q*
0.0
0.1
0.2
0.3
0.4
130
483
457
66
1
2
3
4
Max RF of the pair
(labels: number of pairs)
(d) Geometry level
|Geometry mean residual|
0.0
0.1
0.2
0.3
0.4
1
2
3
4
1367
363
263
53
CF (max in geometry)
number of geometries
(e) Mean |residual| by RF × CF
RF
CF
1
2
3
4
0.30
n = 2503
0.28
n = 181
0.38
n = 322
n = 183
sparse
0.35
n = 2514
0.30
n = 296
0.53
n = 353
n = 11
sparse
0.38
n = 1544
0.27
n = 661
0.55
n = 120
n = 8
sparse
0.42
n = 506
0.60
n = 91
n = 30
sparse
n = 10
sparse